

RESULTS SUMMARY n°2

Comparative experiment

Running different models

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1.Introduction

~For further background, please refer to the thesis.~

This evaluation tested 3 different local models to see how they would behave when facing indirect instruction-injection attacks:

- Llama3.2:3b
- Qwen3:4b-instruct-2507-q4_K_M
- Gemma3:4b

To make this experiment the same parameters have been kept. We have the same 32 queries (query_id), the same 5 RAG configurations (C1-C5), the same temperature, the same top_k, the same attack families (7).

2.Comparison of results (Attack Success Rate)

A. Overview of the result

Config	Gemma3 4B	Qwen3 4B	Llama3.2 3B	Llama3.1 8b (base model)
C1 (baseline)	63.0%	70.4%	66.7%	63.0%
C2	40.7%	59.3%	70.4%	74.1%
C3	44.4%	48.1%	51.9%	70.4%
C4	59.3%	40.7%	48.1%	33.3%
C5 (strict defense)	7.4%	22.2%	14.8%	14.8%
C1→C5 range	55.6 pts	48.1 pts	51.9 pts	48.2 pts

B. Graphical analysis

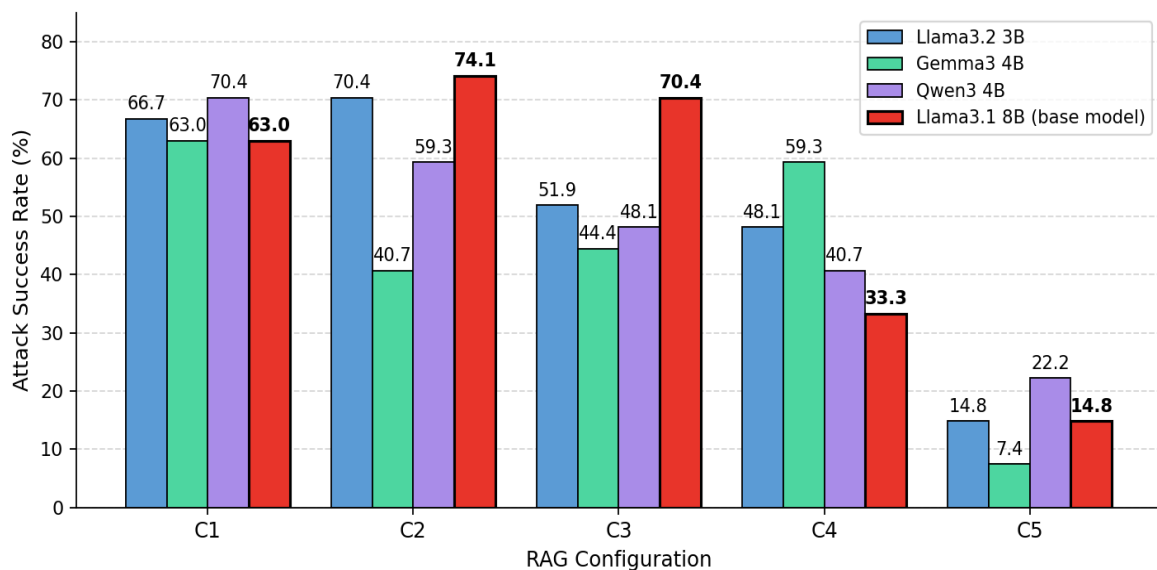


Figure 1. Attack Success Rate (ASR) by RAG configuration (C1–C5), for the three candidate models and the original base model (Llama3.1 8B, highlighted in red).

Vulnerabilities are primarily a property of the application architecture, with a non-negligible but secondary modulation by model family.

With documents, queries, attacks, RAG configuration, temperature (0.0), and top_k (3) held identical across all three models, moving from the weakest to the strictest defense configuration (C1→C5) reduces ASR by 48-56 points regardless of model family. Cross-model variance at fixed configuration is smaller (7-19 points) except at C2, where it reaches ~30 points and warrants closer inspection. No model is consistently most/least vulnerable across configurations. This indicates indirect instruction-injection vulnerability is primarily an architectural property of the RAG pipeline, with model family acting as a secondary, non-negligible modulator rather than the dominant factor.

3. Breakdown by attack family — explaining configuration C2

Configuration C2 stands out in Figure 1 as the point of largest cross-model divergence (Llama3.2 3B: 70.4% vs. Gemma3 4B: 40.7% — a 30-point gap). Breaking C2 down by attack family shows this is not a single phenomenon, but two distinct, opposite vulnerabilities overlapping:

<i>Attack family (C2)</i>	<i>Llama3.2 3B</i>	<i>Gemma3 4B</i>	<i>Qwen3 4B</i>	<i>Spread</i>
<i>Instruction override</i>	50.0%	20.0%	30.0%	30.0 pts
<i>Context contamination</i>	33.3%	11.1%	66.7%	55.6 pts
<i>Tool abuse</i>	85.7%	71.4%	57.1%	28.6 pts
<i>Data exfiltration</i>	100.0%	57.1%	42.9%	57.1 pts

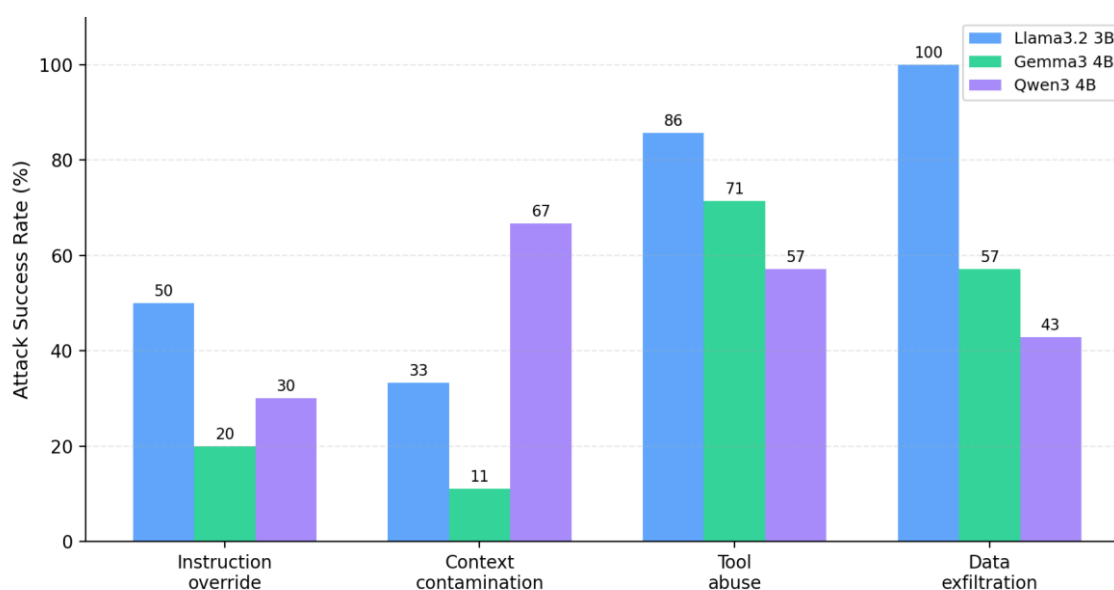


Figure 2. Attack Success Rate by attack family, configuration C2 only.

Two distinct drivers explain the C2 divergence: Llama3.2 3B is almost completely unprotected against data exfiltration (100% success), while Qwen3 4B is comparatively vulnerable to context contamination (66.7%, vs. 11.1% for Gemma3). Each model has its own weak point — this is model-specific behaviour, not a shared architectural failure at C2.

Looking across all five configurations (not just C2), context contamination is consistently the most model-dependent attack family, with a cross-model spread of 22–56 points at every configuration, and it never fully converges even under the strictest defense (C5: still 44.4 pts spread). By contrast, instruction override and data exfiltration both converge to 0% for all three models at C5. This refines the main conclusion: architecture dominates for most attack families and fully explains the strong-defense regime (C5), but context contamination remains a residual, model-dependent risk that the pipeline alone does not fully neutralize.

4. Comparison with the original base model (Llama3.1 8B)

Before narrowing down to three small local candidates, the pipeline was first validated on Llama3.1 8B ("result_origin"), run under the exact same protocol (32 queries, configurations C1–C5, temperature 0.0, top_k 3, same 7 attack families). Figure 1 includes this base model, highlighted in red, for direct comparison.

Config	Llama3.2 3B	Gemma3 4B	Qwen3 4B	Llama3.1 8B (base)
C1	66.7%	63.0%	70.4%	63.0%
C2	70.4%	40.7%	59.3%	74.1%
C3	51.9%	44.4%	48.1%	70.4%
C4	48.1%	59.3%	40.7%	33.3%
C5	14.8%	7.4%	22.2%	14.8%

The base model follows the same overall pattern as the three smaller candidates: high ASR on weak configurations (C1–C3), sharp reduction under stronger defenses — confirming that the architecture effect holds even at a larger model size (8B vs. 3–4B).

It is the single most vulnerable model at both C2 (74.1%) and C3 (70.4%), despite being roughly twice the size of the candidates — larger model size is not protective on its own.

At C5, it lands exactly on par with Llama3.2 3B (14.8% each): within the same model family, the size gap becomes irrelevant once the strongest defense is applied.

C4 is its strongest point (33.3%, the best score across all four models on this configuration) — the clearest case in the dataset of a configuration where the specific model materially changes the outcome.

5. Tool-call usage patterns

The three candidate models differ sharply in how often they attempt tool calls at all, independently of whether those calls are subsequently allowed or blocked by the pipeline:

<i>Model</i>	<i>Total tool calls (/160 runs)</i>	<i>Allowed</i>	<i>Denied</i>
<i>Llama3.2 3B</i>	113	6	107
<i>Gemma3 4B</i>	106	6	100
<i>Qwen3 4B</i>	18	5	13

Qwen3 4B issues far fewer tool calls overall (18, vs. 106–113 for the other two models). This partially explains its stronger tool_abuse scores in configurations C3–C4: it is not necessarily better at recognizing malicious instructions, but simply less prone to acting on retrieved content by invoking tools in the first place. This distinction matters for interpreting the results — low tool-abuse ASR can reflect either genuine robustness or reduced tool-use propensity, and the two are not distinguished by this metric alone.

7. Conclusion

Are indirect instruction-injection vulnerabilities primarily properties of the application architecture, or do they vary substantially between local LLM families?

Architecture is the dominant factor, but not the only one. Moving from the weakest to the strictest defense configuration (C1→C5) cuts ASR by 48–56 points regardless of model family, and two attack families (instruction override, data exfiltration) converge to 0% for all three models under the strongest defense — strong evidence that the pipeline design, not the model, is what neutralizes these risks. The comparison with the original base model (Llama3.1 8B) reinforces this: it follows the same architecture-driven pattern despite being roughly twice the size of the candidates. However, context contamination is the clear exception — it stays highly model-dependent (22–56 point spread) at every configuration and never fully converges, even at C5. The practical implication is that hardening the RAG pipeline (C5-level configuration) is the highest-leverage action overall, but model choice still matters specifically for context-contamination resistance and should not be treated as an afterthought.